



An analysis of sonde data from the Garren at Packsaddle Bridge

A report prepared on behalf of Wye Salmon Association (WSA)

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<i>Author</i>	<i>Gordon Green</i>

Introduction

This report reviews the data output from a collaboration between the WSA citizen science team and Dr. Liz Bagshaw from the University of Bristol. On 18th August 2025, a multi-sensor HydroVu sonde was deployed on the Garren Brook, just upstream of Packsaddle Bridge. The objectives for WSA were (a) to evaluate the general utility of sonde deployment on a eutrophic stream in an area of intensive, mixed agriculture, and (b) to look specifically at nitrate and ammonia to check for transient events which might be missed by weekly CS monitoring. The sonde has now been in place for five months, so a review of the results is appropriate.

The primary sonde included sensors for water temperature, depth, turbidity, electrical conductivity, nitrate and ammonium. pH was not measured, to make space for both nitrate and ammonium which are of particular interest at present due to their role as eutrophication drivers. Unfortunately, the nitrate and ammonium sensors both have a high range, more appropriate to sewage effluent than to surface water. Thus, it was foreseen that we would probably need to cross-calibrate with standard CS measurements. A secondary sonde, measuring temperature and dissolved oxygen was also deployed, however the oxygen sensor was found to be faulty and has yielded no useful information. Consequently, this parameter will not be considered further.

The sonde was deployed within a protective tube (drainpipe) anchored to a stake driven into the gravel substrate. The top of the protective tube was fixed to a bankside tree, with the sonde's telemetry unit being fixed in the branches above the bank. The sonde location was in on the inside of a gentle bend in circa 250mm of flowing water (at low summer flows). The deeper water on the outside of this bend is a known fish holding area (mainly chub). Approximately 50m downstream of the sonde, Packsaddle Bridge is a regular WSA water quality monitoring site (Ref. WSA125 on EpiCollect/WyeViz). A few hundred metres upstream is the EA flow monitoring site at Marstow Mill.

Results

Temperature

The temperature data, reproduced in figure 1, show the expected seasonal variation. The highest temperature recorded was 17.2°C, the lowest 4.2°C. The fuzziness seen on the trace during the early months derives from the diurnal solar heating cycle which becomes negligible in the cooler months. Figure 2 shows the cycle during one day in August. The CS results track the sonde very well at higher temperatures but tend to be a degree or so high at lower temperatures.

Temperature measurement is clearly a robust and stable parameter in the context of remote sensing.

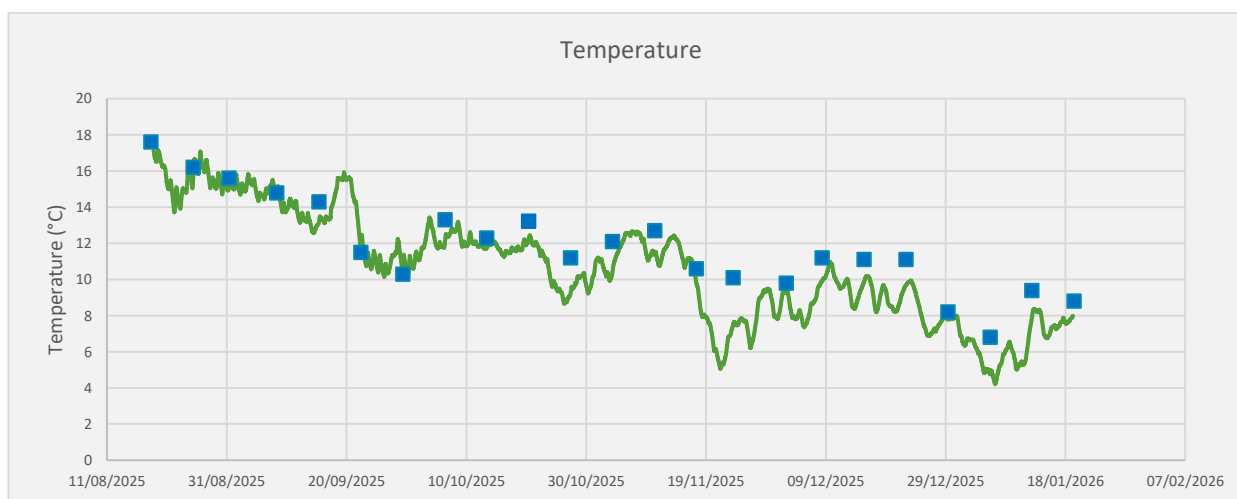


Figure 1: temperature data (green); blue squares are the CS measurement at WSA125.

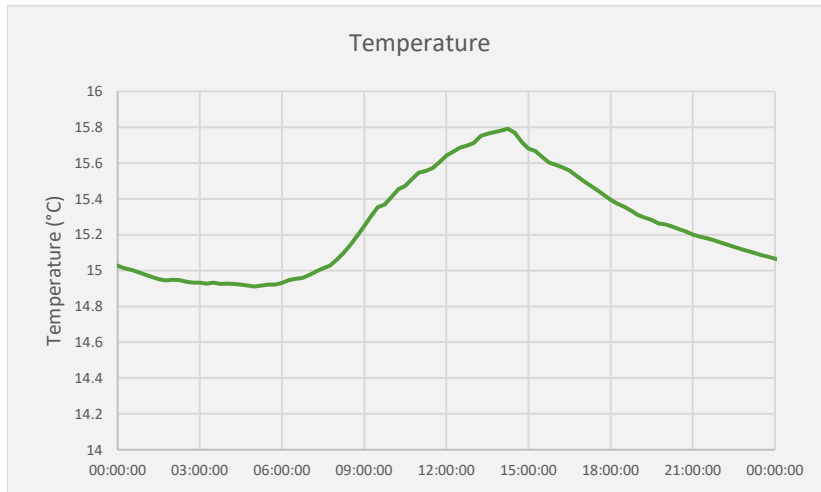


Figure 2: diurnal temperature cycle on 31st August (looks like it clouded over mid afternoon!).

Depth

Depth is measured as the difference in pressure between a sensor within the sonde and a sensor in the bank-side telemetry unit (atmospheric pressure), both of these parameters being separately available. Figure 3 shows the five-month depth trace.

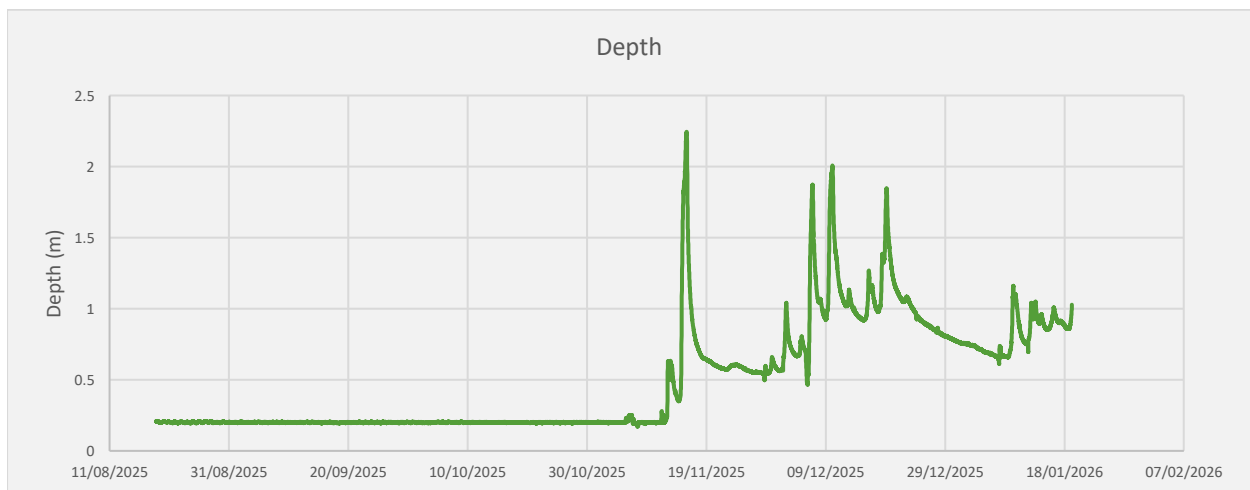


Figure 3: depth data

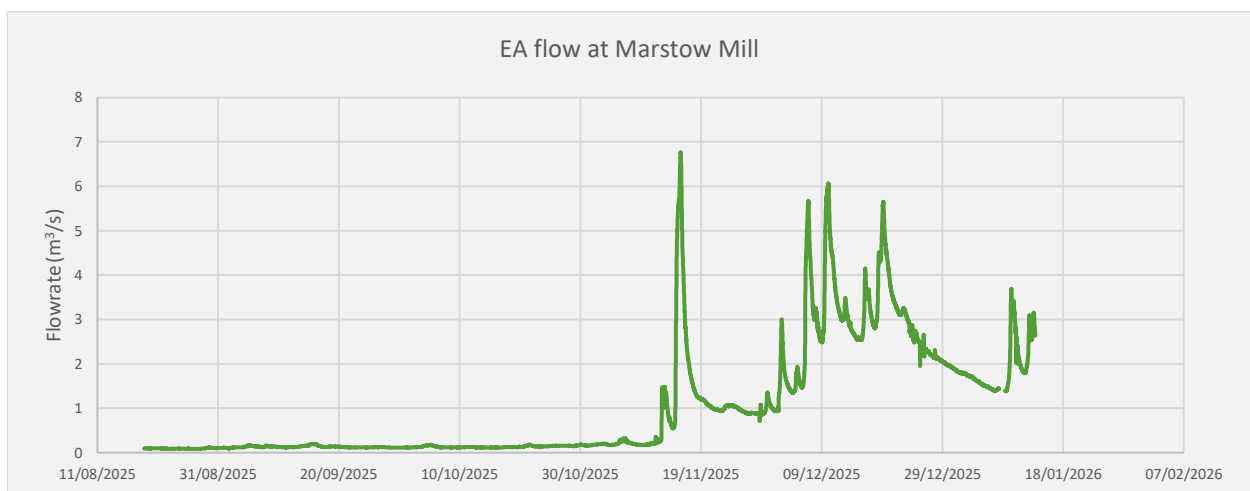


Figure 4: corresponding volumetric flowrate recorded at Marstow Mill

After a dry summer, the river was on its bones until November. Overall, the depth trace matches the volumetric flow data very well (the two are related in a non-linear fashion, so correspondence will not be exact). Again, we appear to have a robust parameter with no obvious drift or drop-outs.

Specific conductivity

Specific conductivity is a two-parameter measurement. First, the probe measures electrical conductivity. However, because conductivity is known to be temperature sensitive, readings are usually referred back to 25°C using the measured water temperature (this calculated 25°C value being the specific conductivity). Figure 5 shows the SC trace and corresponding CS measurements.

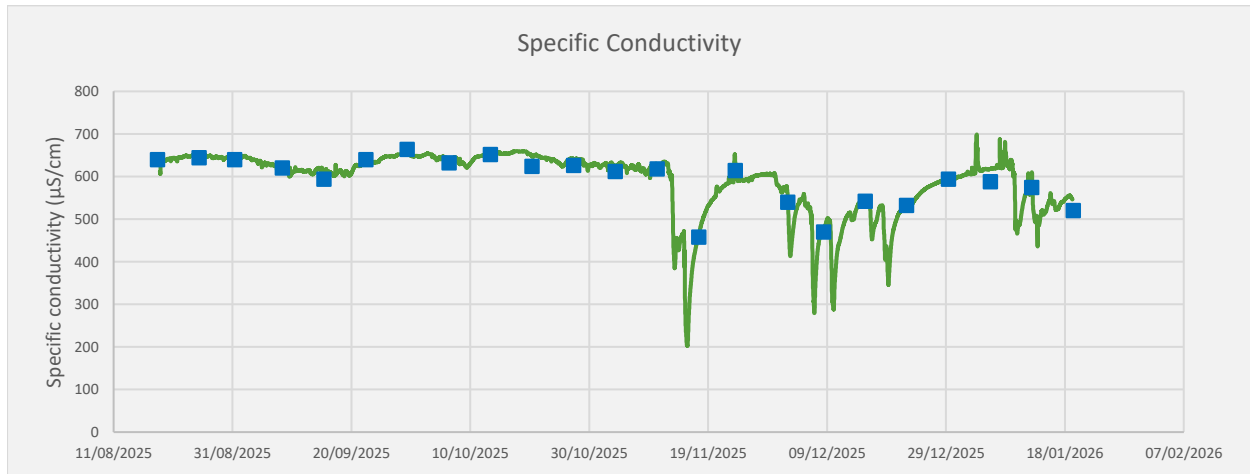


Figure 5: specific conductivity data (green); blue squares are the CS measurement at WSA125.

Low flow values are exactly where we expect them to be in this high alkalinity stream. The downward transients correlate with the high-flow periods seen in figure 3 and 4 and are exactly as expected. CS data from WSA125 match this trace perfectly. The four upward transients, one in November and three in January are of interest. This is the type of stuff we do not capture with weekly sampling. Moreover, specific conductivity appears to be a simple and robust technology, exhibiting no issues over the five-month deployment.

Turbidity

The turbidity trace is shown in figure 6, again alongside CS data from WSA125.

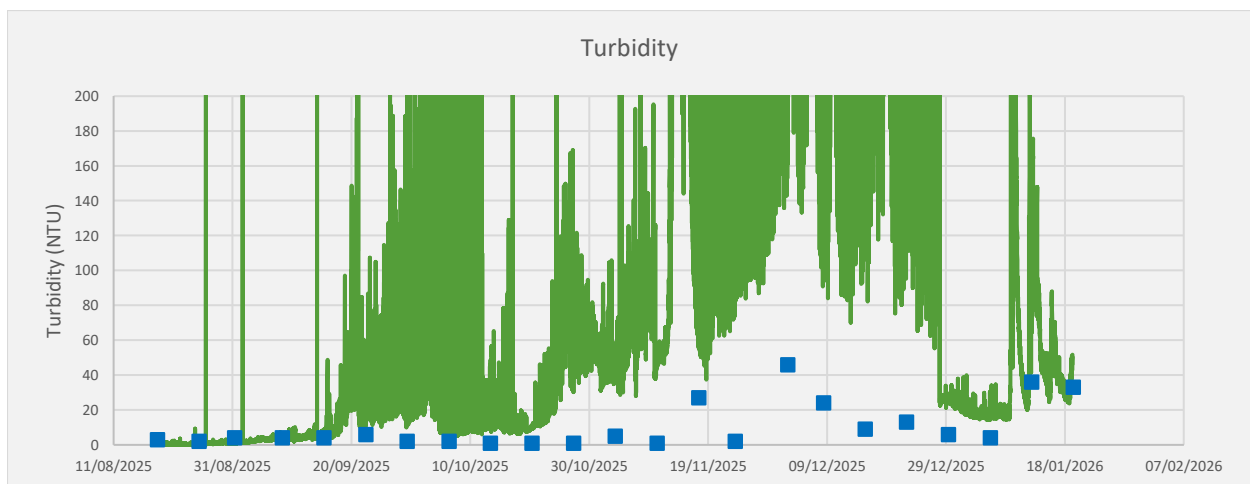


Figure 6: turbidity data (green); blue squares are the CS measurement at WSA125.

The sonde data tracks the CS data reasonably well for the first month, excepting two transient spikes. But from 20th September onwards, we clearly have major contamination trapped in front of the sensor. Peak readings are high – several thousand NTU. This represents optical blockage, not turbidity; a

problem not infrequently seen on EA sondes, despite their built-in wipers. I imagine EA despatch someone to withdraw the sonde and rinse it when this occurs. But for CS deployment, this option is not so simple.

Ammonium

The ammonium trace is shown in figure 7, alongside WSA Hanna “ammonia” data (actually ammoniacal N) from WSA125.

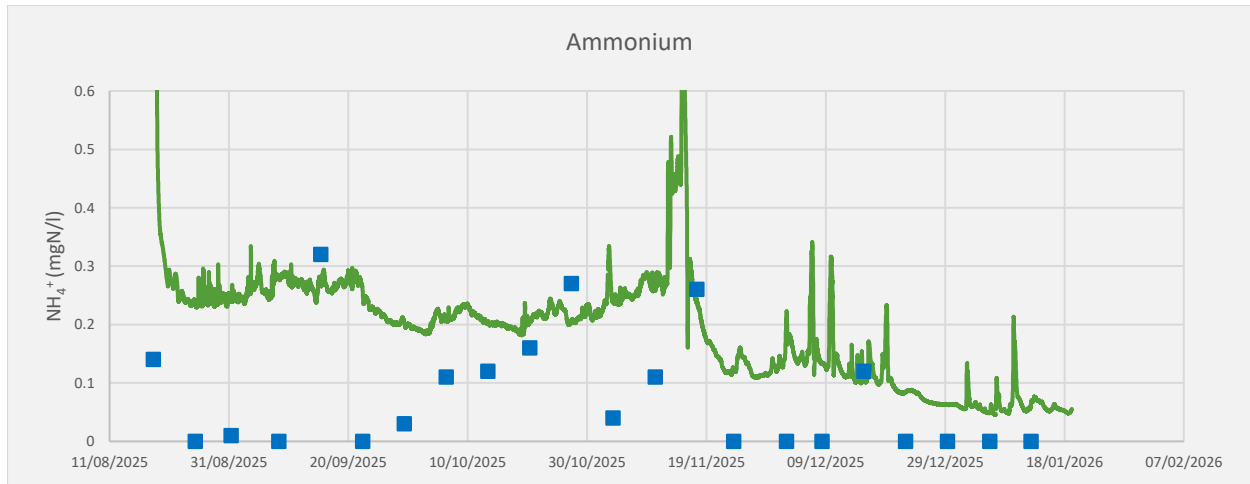


Figure 7: ammonium data (green); blue squares are the CS measurement at WSA125.

There is clearly a serious zero drift taking place on this sensor. Whilst this could be mathematically compensated (at least approximately) in retrospect, it does make real-time monitoring of absolute readings problematic. Drift is a known issue with all ion-selective membrane type sensors and occurs on EA sondes too. EA minimise this by swapping out sondes, typically every three months, and recalibrating. Despite this, EA sondes seem to exhibit a poor correlation to laboratory determinations from proximate locations. In a qualitative sense, though, we can clearly identify large ammonium spikes during high flow events (comparing to figure 3), especially the first of the winter around 12th November, also 6th & 10th Dec. There are also some smaller spikes evident (e.g., 1st January) which do not have obvious flow correlations. So, as for conductivity, we are picking up some transient signals which are not captured by weekly monitoring. Nevertheless, this is arguably a troublesome technology for wider CS deployment.

Nitrate

The nitrate trace is shown in figure 8, alongside WSA test strip data (TON) from WSA125.

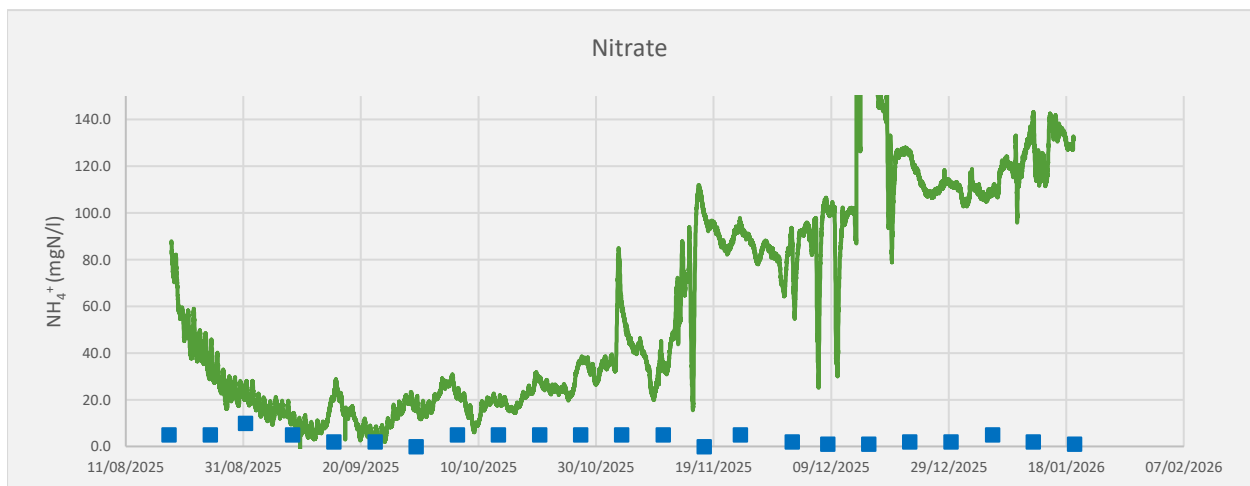


Figure 8: nitrate data (green); blue squares are the CS TON measurement at WSA125.

The nitrate sensor (another ion selective membrane) took three weeks to settle down, then proceeded to drift upwards for the remainder of the deployment. The same comments apply as for ammonium; there is no simple way to turn this into good quantitative data. In a qualitative sense, the three larger negative spikes in November and December, correlate with positive spikes on ammonium and with flow spikes seen in figure 3 & 4. But overall, this is again not a very straightforward technology for CS application.

Conclusion

- Temperature, depth and electrical conductivity are all straightforward, stable and robust sensor technologies which would suit CS deployment. More widespread deployment of telemetric monitoring of these parameters could be considered for the future, particularly in the context of low cost IOT solutions. This could potentially offer some labour saving for volunteers, or the possibility of monitoring where we do not have volunteers.
- Turbidity, necessarily an optical measurement, will always be a difficult technology for long-term deployment in rivers.
- Nitrate and ammonium sensors have also proved troublesome in this deployment. Although the ranges of these particular sensors were known to be inappropriate, the zero drift problems encountered are typical of ion-selective membranes in general. Indeed, WSA has experienced this with the HM Digital pH meters. High end sondes often substitute optical methods to achieve more stable measurement of substances such as nitrate. However, this is expensive technology.
- Within the dataset examined, transient events can indeed be observed on electrical conductivity, nitrate and ammonium traces. Some of these correlate with flow spikes, some do not. Note that nutrient spikes during high flow events are not discernible on electrical conductivity because the effect is overwhelmed by the massive fall in mineral ions occurring at such times. More generally, electrical conductivity will never show up typical variation in nutrient levels; even nitrate concentration is two orders of magnitude down on dissolved minerals. Nevertheless, electrical conductivity would be capable of providing an indication of gross digestate spills and the like.

Acknowledgements

Thanks are due to Dr. Liz Bagshaw for providing the sonde and for supporting its installation and set-up. Also, thanks to Peter Hunt, a stalwart of the WSA CS community, for organising bank access and for his diligent collection of water quality data.